

MOHAMED ADEL REZK M.Sc.

Principal Data Engineer (ML/AI)

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EDUCATION

M.Sc. in Management Information Systems

📅 Sept 2015 – June 2017

Alexandria University & National University of Ireland, Galway

- Graduate research in **Linked Data** and coursework in **Distributed Systems**, and **Information Systems Architecture**, supporting foundational software engineering and data-intensive system design.
- M.Sc. Dissertation: A Government Support System Based on Citizen Interactions**
Proposed and implemented a framework for **linked public policy management**, combining semantic data modeling, NLP, sentiment analysis, and citizen-interaction signals to support evidence-based decision making.

B.Comm. in Management Information Systems

📅 Sept 2006 – June 2010

Alexandria University

- Studied **Database and Data-warehouse Design**, **OOP (Java & C#)**, **Networking**, **Systems Design**, and **Mathematics**, building early competency in software engineering and system modeling.

SKILLS

Programming	Python, SQL, Java, Scala, Bash, JavaScript, C#, NoSQL, SPARQL, LaTeX
Cloud & Platforms	Microsoft Azure (ADF, Synapse, Databricks, Functions, WebApps, Data Lake, Key Vault, ACI, ACR), Snowflake
Data Engineering	ETL/ELT Pipelines, Distributed Processing, Apache Spark, Databricks, Delta Lake, Data Modeling, Data Warehousing, Schema Evolution, Data Contracts, Data Quality (DBT, Great Expectations, Schema Change)
DevOps & Automation	CI/CD (GitHub Actions, Jenkins), Terraform (IaC), Deployment Automation, Release Management, Environment Promotion, Rollback Strategies
Reliability & Security	Reliability Engineering, Observability (logging, metrics, tracing), Runbooks, BCDR, RBAC, Secrets Management, Data Masking, Unity Catalog Governance
ML & Analytics	Applied Data Science, Time-Series Analysis, Anomaly Detection, Statistical Modeling, Feature Engineering, Model Deployment (FBProphet, scikit-learn)
NLP & Semantic	NLP, Named Entity Recognition (NER), Semantic Web, Linked Data, RDF/JSON-LD, Knowledge Graphs, Ontology Design, SPARQL, Data Cube Vocabulary
Tools & Ecosystem	Git, Jupyter Notebook, Airflow, Docker, Nginx, Hadoop, MongoDB, Elasticsearch, CKAN, Tableau
Architecture & Leadership	Cloud Native System Design, Microservices, Platform Engineering, Scalability & Fault Tolerance, Cost Optimization, Cross-Team Collaboration, Technical Leadership, Mentoring, Stakeholder Communication

EXPERIENCE

Principal Data Engineer, TLCP

📍 Dublin, IE

📅 Feb 2021 – Present

OPEN Team — Optum Advanced Technology (UnitedHealth Group)

Projects: EMISAR Analytics Gateway, Enlighten

- Led end-to-end design, implementation, and evolution of a **cloud-native data platform** on Azure, powering (EMISAR) a multi-billion revenue analytics gateway across ingestion, transformation, orchestration, and delivery.
- Authored and owned **High-Level Design (HLD) documentation** for core platform components and major initiatives, aligning architecture, scalability, security, and operational considerations across engineering and stakeholder teams.
- Designed and delivered multiple production-grade **DBT implementations**, leveraging Azure Durable Functions, Azure Container Instances, GitHub Actions, Azure Data Factory, Airflow, and DBT Cloud to support scalable and automated transformations while driving infrastructure and operational costs down.
- Led several **platform, data warehouse, and tooling migrations**, including Synapse → Snowflake, Snowflake V2 → multi-tenant Snowflake architecture, self-hosted Azure Databricks → UDLP tenant, DBT self-hosted → DBT Cloud, Azure Data Factory → Airflow, self-hosted GitHub → GitHub Enterprise, and Scala-based extractors → Python-based services to improve efficiency, maintainability, and cost control.
- Modernized CI/CD by leading the migration from **Jenkins to GitHub Actions**, redesigning pipelines to improve reliability, consistency, and developer experience while reducing operational overhead.
- Designed, developed, and maintained **batch and near-real-time data export APIs** supporting downstream consumers, implemented using DBT, PySpark, Databricks, Delta Lake, Unity Catalog, Java, Python, Azure Functions and Queues, and Snowflake.
- Designed and implemented a **secure, self-managed NocoDB deployment** using Azure Webapp, Containers and SqlServer to enable controlled entry of business-critical contractual data, integrating access controls and audit considerations, and embedding contractual metadata directly into the ETL/ELT pipelines to enrich downstream datasets.

- Planned and executed **multiple successful production releases** using a structured release management approach focused on risk mitigation, validation, and rollback readiness, consistently achieving near-zero production impact.
- Built **scalable ETL/ELT pipelines** using Spark (Databricks), Python, Scala, SQL, Snowflake/Synapse, and Delta Lake; introduced schema evolution and automated data quality enforcement using Flyway, Schema Change, DBT, and Great Expectations.
- Established an **automation-first delivery model** with Terraform IaC, GitHub Actions/Jenkins, reusable deployment templates, and controlled CI/CD pipelines to improve change safety, reproducibility, and team velocity.
- Authored and maintained **BCDR (Business Continuity and Disaster Recovery)** documentation and recovery procedures to ensure service availability, recovery readiness, and operational resilience.
- Secured (EMISAR) using Azure Key Vault, role-based access controls, credential rotation, and data masking to meet security and compliance requirements.
- *Tech:* SQL, Python, Scala, Terraform, Spark/Databricks, Unity Catalog, DBT, Great Expectations, Schema Change, Flyway, GitHub Actions, Jenkins, Azure (ADF, Synapse, Functions, WebApps, Data Lake, Key Vault, ACIs, ACR), Snowflake, NocoDB.

Senior Data Scientist

📍 Dublin, IE

📅 Jul 2019 – Feb 2021

Total Cost of Care Team – OptumRx (UnitedHealth Group)

Project: Drug Utilization Spikes Detection

- Designed, implemented, and automated **statistical anomaly detection system** to monitor drug utilization patterns across multiple business sectors within Total Cost of Care program.
- Built **end-to-end data pipelines** for ingestion, transformation, modeling, and alerting using SQL, Python, and PySpark on Hadoop-based platforms.
- Developed and operationalized **time-series models** (e.g., Prophet) to detect abnormal utilization trends, enabling early identification of drugs utilization risks.
- Integrated analytical outputs with **Tableau dashboards** to communicate insights effectively to clinical, operational, and business stakeholders.
- Collaborated with cross-functional teams to productionize models, automate workflows, and improve reliability of anomaly monitoring processes.
- *Tech:* Python, SQL, Bash, PySpark, FBProphet, Hadoop DFS, IBM Netezza, Tableau (Desktop/Server), Jupyter, Git, Unix/Windows; data formats: CSV, JSON, Parquet.

Senior ML/AI Research Engineer – (Part-time PhD Candidate - On Hold)

📍 Galway, IE

📅 Jan 2018 – Jun 2019

Data and Knowledge Engineering Unit – Insight Centre for Data Analytics (SFI funded)

Project: Automating Data Science – PhD Research ML/AI

- Conceptualized and scoped the **automation of data science workflows**, defining problem taxonomies and modeling approaches for automating analytical task selection and execution.
- Designed, developed, and maintained a **full-stack web application** for structured domain-expert knowledge collection, including UI in HTML and JavaScript, a Django-based web server, MongoDB as the primary datastore, Elasticsearch-backed caching, and Nginx load balancing.
- Built an **analytics and insights portal** on top of the expanding knowledge base, enabling visualization and exploration of domain patterns as the underlying dataset evolved.
- Designed and trained **regression and classification models** to predict **(a) the most suitable data science pipeline** for a given problem and dataset, and **(b) the platform and computational resources** required to achieve optimal performance based on dataset and tasks characteristics.
- Leveraged dataset metadata and statistical features (e.g., size, sparsity, feature distributions, data types, and platform engagement) to inform automated recommendations for modeling approaches and execution environments.
- Designed and implemented **data acquisition and feature-engineering pipelines** by scraping, normalizing, and enriching heterogeneous knowledge sources to support automated model generation.
- Applied **Data collection and cleaning feature engineering and selection, classification, and regression techniques** to predict problem-specific engagement metrics (e.g., Problem Engagement Index).
- Developed a **Python wrapper over high-performance C++ components** (inclose4) to integrate frequent pattern mining algorithms into the automated modeling pipeline.
- Prototyped and evaluated solutions using reproducible experimentation workflows, collaborating with researchers to validate results and iterate on system and data model design.
- *Tech:* Python, JavaScript, Bash, HTML, Django, MongoDB, Elasticsearch, Jupyter, Git; libraries/tools including Nginx, Flask, Greyhound, Cube.js, Selenium, Kaggle API, pymongo, mlxtend, spmf, inclose4; Unix/Windows; data formats: CSV, XML, JSON.

ML/AI Research Engineer

📍 Galway, IE

📅 Oct 2015 – Jan 2018

eGov Unit – Insight Centre for Data Analytics

M.Sc. Dissertation: A Government Support System Based on Citizen Interactions

Proposed and implemented a framework for **linked public policy management**, combining semantic data modeling, NLP, and citizen-interaction signals to support evidence-based decision making.

- Designed and implemented a **Core Public Policy Vocabulary (CPPV)** to standardize representation of public policies and enable interoperability across government datasets and platforms. CPPV can demo.

- Built tooling and pipelines for **Public Policy Collaboration Mining (collaboration miner)**, analyzing metadata from open data catalogues to identify collaboration opportunities between policy stakeholders.
- Developed **Public Policy Acceptance Prediction (GOV-DAF)** by analyzing historical social media interactions related to public policies, enabling early forecasting of citizen response to proposed initiatives.
- Delivered end-to-end prototypes covering data ingestion, semantic modeling, analytics, and visualization, with open-source implementations and published research outputs.
- *Tech:* Python, Java, SPARQL, JavaScript, MongoDB, AWS Lambda, Flask, CKAN; libraries including Twitter API, Stanford NER, DBpedia Spotlight, LDA; data formats: JSON, RDF, OWL, PDF.

Project: European Linked Open Statistical Data Hub (LOSD)

- Consulted with external partners (Derilinx, ADAPT Centre) on the adoption of **RDF Data Cube vocabulary** standards for publishing interoperable statistical data.
- Co-designed and implemented the **Linked Open Statistical Data (LOSD) pipeline**, enabling ingestion, transformation, and publication of heterogeneous statistical datasets as linked data.
- Built and integrated data processing components using Python, Java, and SPARQL, supporting scalable data modeling, validation, and visualization workflows.
- Co-organized and supported a **multi-national testing hackathon** with European statistical offices at INSEE (Paris), collaborating with domain experts to validate data pipelines and standards compliance.
- *Tech:* Python, Java, SPARQL, JavaScript, RDFLib, Pandas, NumPy, Matplotlib, Django, Docker, Virtuoso, Nginx, Git; data formats: RDF, JSON-LD, XML, CSV.

Project: Open Government Intelligence

- Designed and implemented a suite of **data publishing and transformation tools** to convert heterogeneous government datasets into interoperable Linked Data using RDF Data Cube standards.
- Built and open-sourced multiple platform components, including a **Cube Builder**, **Schema Creator**, **real-time RDF publishing pipeline**, **Table2QB Python wrapper**, and country-specific data restructuring pipelines (e.g., Lithuanian pilot).
- Enabled **real-time and batch data processing workflows** for analytics and monitoring use cases, including renewable energy planning, search and rescue operations, and business KPI modeling.
- Developed data ingestion, modeling, and publication pipelines using Python, Java, SPARQL, and JavaScript, integrating semantic storage, validation, and visualization layers.
- Collaborated with cross-functional and international stakeholders to validate pipelines, schemas, and operational workflows across multiple public-sector pilots.
- *Tech:* Python, Java, SPARQL, JavaScript, Clojure, GraphQL; AWS Lambda, Django, Druid/Superset; RDFLib, Pandas, NumPy, scikit-learn; Virtuoso, Docker, Nginx, Git; data formats: RDF, JSON-LD, XML, CSV.

Project: ROUTE-TO-PA

- Designed and implemented **data quality assessment and anomaly detection services** for datasets published on CKAN-based open data platforms.
- Built an **unsupervised anomaly detection pipeline** using Local Outlier Factor (LOF) to identify data inconsistencies, outliers, and quality issues in published datasets.
- Developed lightweight **Python-based services** and APIs using Flask, enabling automated validation and quality checks as part of data publication workflows.
- Integrated statistical analysis and visualization to support dataset inspection and issue triage for platform operators and data publishers.
- *Tech:* Python, Flask, CKAN, Git; libraries including pandas, NumPy, scikit-learn, matplotlib, pylof; data formats: JSON.

Project: Your Data Stories

- Implemented **batch data processing and restructuring pipelines** for Galway traffic datasets, preparing heterogeneous data for RDF-based publication and downstream analytics.
- Designed and executed data transformation workflows to normalize, enrich, and convert raw CSV/JSON inputs into structured, machine readable formats suitable for semantic publishing.
- Supported data exploration and validation through iterative analysis and visualization during development.
- *Tech:* Python; data formats: CSV, JSON; tools and environments: PyCharm, Jupyter Notebook, Unix.

NLP Research Engineer

📍 Galway, IE

📅 Jan 2015 – Oct 2015

NewsLab - Insight Centre for Data Analytics

Project: Social Repository of Ireland

- Designed and implemented a **semantic social media repository** as part of Digital Repository of Ireland (DRI), aggregating, enriching, linking, and visualizing Twitter data related to Irish national events.
- Built data ingestion and enrichment pipelines to collect social media streams, apply entity recognition and semantic similarity techniques, and persist enriched datasets for analysis and visualization.
- Enabled cross-dataset linking and exploration by combining social signals with structured metadata to support research and archival use cases.
- *Tech:* Java, Node.js, JavaScript, MongoDB; libraries including Twitter API, Stanford NER, DISCO semantic similarity; tools: Maven, NPM, Git; Unix environments.

Project: Newswire 2.0

- Built a **news ingestion and analytics pipeline** that collected articles from RSS feeds and analyzed their popularity using Named Entity Recognition (NER) signals derived from social media activity.
- Developed models to compute entity-level popularity indices by correlating news content with social media sentiment and engagement trends.
- Implemented end-to-end workflows covering data collection, enrichment, storage, and analysis for large-scale news datasets.
- *Tech:* Java, JSON, XML, MongoDB; libraries including Twitter API and Stanford NER; tools: Maven, Git, Unix.

Teaching Assistant (ML/AI)

📍 Alexandria, Egypt

📅 May 2011 – Dec 2014

Department of Computers and Information Systems, Alexandria University

- Delivered hands-on laboratory instruction for **Systems Analysis and Design, Object-Oriented Programming, Database Design, e-Commerce, and Artificial Intelligence** (Neural Networks, Fuzzy Logic).
- Supervised undergraduate capstone projects, providing technical guidance on system design, implementation, and documentation.

Oracle Applications Support Engineer

📍 Alexandria, Egypt

📅 Nov 2010 – May 2011

ICTP Project, Faculty of Engineering, Alexandria University

- Maintained, extended, and administered university **MIS applications and servers**, supporting SQL-based backends and Java applications in Unix environments.
- Troubleshoot production issues, applied system updates, and ensured availability and correctness of academic and administrative systems.

MOST PROUD OF

Awards and Recognition



CSS Egypt Award

Enhancing Internet User's Experience using Semantic Web Technologies [Enhanca App](#)



Distinction-class honours

Top of 2010 Class (ranked 1st)



TLCP member @UHG

A technical community recognizing high-impact individual contributors.

EVENTS & CERTIFICATIONS (SELECTION)

- **Open Government Intelligence Plenary Meetings** — CERTH (Thessaloniki), Enterprise Lithuania (Vilnius), Marine Institute (Dublin)
Demos, technical presentations, and cross-organization collaboration on data publishing pipelines and standards.
- **LOSD Publishing Pipeline Hackathons** — INSEE (Paris), CSO (Dublin)
Hands-on testing, validation, and iterative design of large-scale statistical data publishing pipelines.
- **ICE-GOV Conference 2018** — NUI Galway
Poster and paper presentation on open government data, analytics, and semantic technologies.
- **AI & the Smart Enterprise Event 2018** — Dublin City University
Poster and panel participation on applied AI and data-driven enterprise systems.
- **Deep Learning Workshop 2017** — Insight Centre, DCU
Hands-on training covering neural networks and applied deep learning concepts.
- **Blockchain Workshop 2018** — Insight Centre, NUI Galway
- **Tableau Workshop** — Insight Centre, NUI Galway
- **Advanced Java Programming (Oracle)** — ITI, Alexandria
Certification in advanced Java programming concepts and enterprise development.

PUBLICATIONS



Thesis

- Rezk, Mohamed Adel (Aug. 2017). *A Government Support System Based on Citizens' Interactions: A Comparative Study of Different Arabic Language Analysis Methodologies*. URL: github.com/mohadelrezk/resume/raw/master/publications/MSc_Dissertation_MohamedAdel_Final_August2017.pdf.



Conference Proceedings

- Rezk, Mohamed Adel, Adegboyga Ojo, et al. (2018). "A Predictive Government Decision Based on Citizen Opinions: Tools & Results". In: *Proceedings of the 11th International Conference on Theory and Practice of Electronic Governance*. ACM, pp. 712–714.
- Stasiewicz, Arkadiusz et al. (2018). "Using Linked Statistical Data to Improve Marine Search and Rescue Operations in Ireland". In: *Proceedings of the 11th International Conference on Theory and Practice of Electronic Governance*. ACM, pp. 412–418.

- Rezk, Mohamed Adel, Mahmoud H Aliyu, et al. (2017a). "Standardising Public Policy Documentation to Foster Collaboration Across Government Agencies". In: *Working Conference on Virtual Enterprises*. Springer, pp. 466–477.
- – (2017b). "Towards a Vocabulary for Open Public Policies". In: *Proceedings of the 10th International Conference on Theory and Practice of Electronic Governance*. ACM, pp. 584–585.
- Rezk, Mohamed Adel, Adegboyega Ojo, and Islam A Hassan (2017). "Mining Governmental Collaboration Through Semantic Profiling of Open Data Catalogues and Publishers". In: *Working Conference on Virtual Enterprises*. Springer, pp. 253–264.
- Rezk, Mohamed Adel, Ghada A El Khayat, et al. (2016). "A Government Decision Analytics Framework Based on Citizen Opinion". In: *Proceedings of the 9th International Conference on Theory and Practice of Electronic Governance*. ACM, pp. 27–30.

REFEREES

Available On Request.